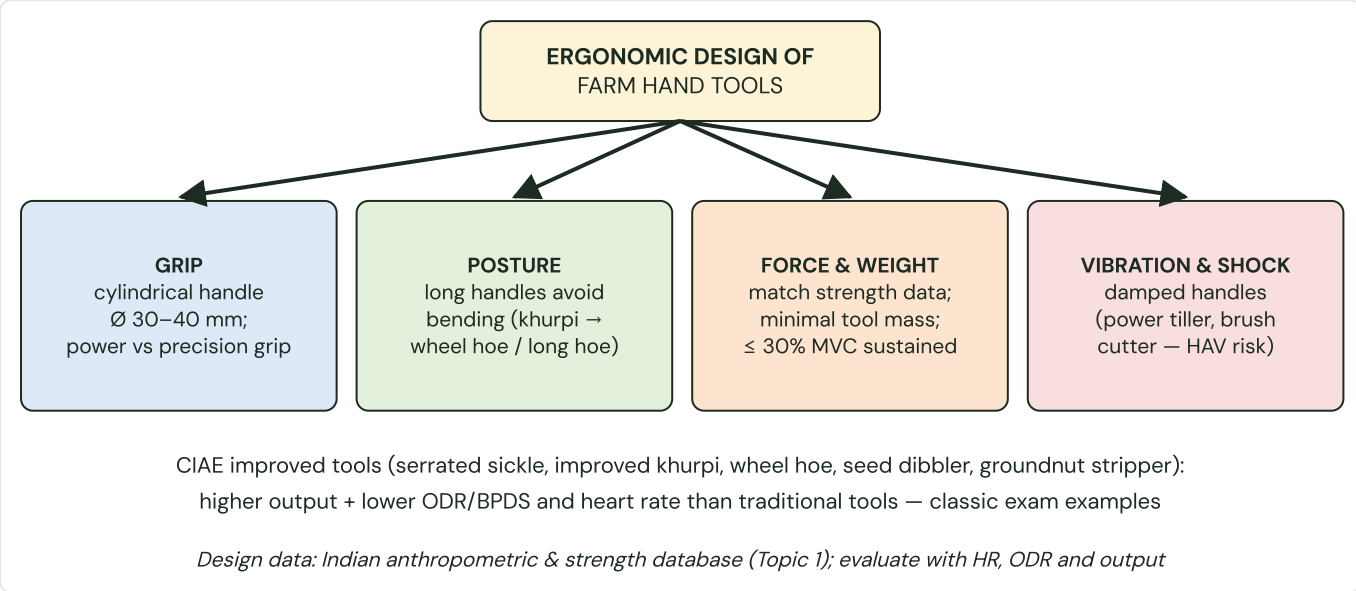
UNIT 5 · TOPIC 4

Ergonomics in Design of Farm Tools

Syllabus topic: Ergonomics in design of farm tools



Ergonomic design of farm hand tools (redrawn from the source figure)

Five design principles

1. **GRIP** — cylindrical handles ~**30–40 mm** diameter suit the power grip (precision tasks use slimmer pen-type grips); smooth, slightly compressible, non-conductive material; no sharp pressure points.
2. **POSTURE** — avoid sustained bending/squatting: long-handled tools (wheel hoe vs khurpi) cut spinal load and energy cost; keep work height near elbow level.
3. **FORCE** — match required forces to Indian strength data (Topic 1); sustained force ≤ ~**30%** of maximum voluntary contraction; use larger muscle groups (legs > arms).
4. **WEIGHT & BALANCE** — minimal tool mass, CG close to the hand.
5. **SHOCK/VIBRATION** damping on powered hand tools (hand-arm vibration: Topic 6b).

Evaluating an improved tool

Follow Unit 3-style testing, but measure the HUMAN as well:

- Output / capacity.
- Heart rate / energy cost.
- ODR / BPDS discomfort.
- Safety observation.

CIAE improved tools — serrated sickle, improved khurpi, wheel hoe, twin-wheel hoe, seed dibbler, cotton-picking bag, groundnut stripper — typically raise output **15–30%** while lowering physiological cost (standard exam examples).

Practice MCQs

Q66. Recommended diameter of a power-grip cylindrical tool handle is:

(a) 10–15 mm
(b) 30–40 mm
(c) 60–80 mm
(d) 100 mm

Q67. Long-handled weeders (wheel hoe) mainly reduce:

(a) Cost
(b) Bending posture and spinal/energy load
(c) Weight
(d) Blade wear

Q68. Ergonomic evaluation of an improved tool measures:

(a) Output capacity
(b) Heart rate/energy cost
(c) ODR/BPDS discomfort
(d) All of these

Q69. CIAE improved hand tools typically achieve:

(a) Lower output
(b) Higher output at lower physiological cost
(c) Same everything
(d) Only lower cost

Answers

Q66 — (b) 30–40 mm
Q67 — (b) Bending posture and spinal/energy load
Q68 — (d) All of these
Q69 — (b) Higher output at lower physiological cost

Sources: Liljedahl et al. — Tractors and their Power Units, Ch. 9 (noise pp. 207–211; vibration pp. 210–218; seating pp. 219–226); ISO 2631; OSHA criteria; CIAE Bhopal improved-tool studies; NPTEL ergonomics lectures.